

Workload Planning

Laboratory for the Information Systems course
in the degree program
Software Engineering and Media Computing

Milestone 2: System Analysis

submitted by

Team 4

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Figure 1: System Boundary4

1 System Boundaries

To gain a clear understanding of the system's scope, the functions to be included and excluded were discussed and visualized, as shown in figure 1.

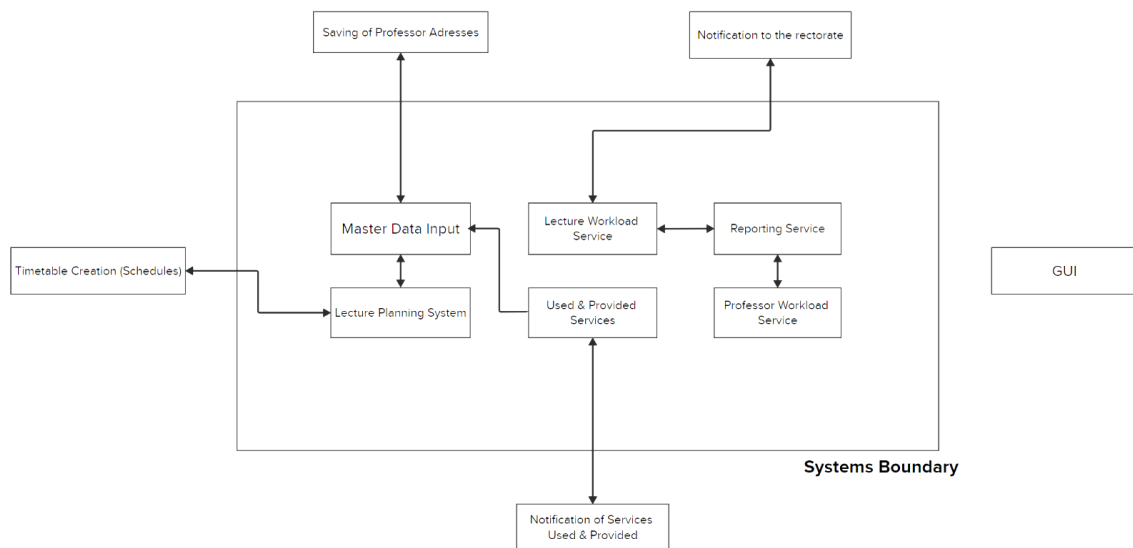


Figure 1: System Boundary

Functions such as timetable creation or sending notifications to others are outside the scope. Additionally, professor addresses are not to be stored in the system, unlike lecturer addresses. The graphical user interface is also excluded from implementation in this laboratory.

The functions within scope will be designed and implemented in the upcoming milestones.

2 User Views

To simplify the development of the requirements later on, major user views were identified and specified while thinking about different application areas.

Import Data View: import the initial data for the information system

- Provides an option to upload the data containing subjects, professors, department information and more

Subject Creation View: input subject information

- Select subject with the corresponding subject number and faculty
- Set the number of weekly hours according to the StuPo and schedule
- Add a remark if necessary

Semester-Specific Course Planning View: plan the lectures, tutorials, and labs offered per degree program and semester and input the workload

- Select a degree program and semester
- Input the courses, professors and lecturers planned for the semester
- Indicate subjects that are offered more or less often due to student volume
- Input the actual number of weekly hours according to the StuPo and schedule after the semester is done

Workload Planning View: manage the weekly hours of professors

- Provide insight into the weekly hours and the credit hour account
- Adjust workload based on tutorials, labs, or shared teaching responsibilities
- Manage additional workloads like functions and the corresponding workload change

Service Creation View: input service information

- Select subject with the corresponding subject number and faculty
- Set the number of weekly hours according to the StuPo and schedule
- Input which faculty profits from this service
- Add a remark if necessary

Service Planning View: plan the lectures offered and used by other faculties and input the workload

- Input the courses used as services, the professors and lecturers
- Input the actual number of weekly hours according to the StuPo and schedule

Report Generation View: create the reports that will be sent out

- List professors' planned number of hours per week and their assigned functions
- List lecturers planned number of hours per week. Include contact information
- List services provided by other departments
- List services used by other departments
- List subjects broken down by degree program and semester

3 Requirements

Lastly, the requirements were engineered while considering the established system boundaries and views.

3.1 Functional Requirements

- The system shall provide an option for users to upload data files containing subjects, professors, department information, and other relevant data
- The system shall validate the uploaded data for consistency and completeness
- The system shall provide access to all tables where imported data needs to be inserted
- The system shall allow the user to see all subject data (subject number, subject name, faculty) in order to create the subjects
- The system shall allow users to set the planned number of weekly hours for each subject based on the StuPo and schedule
- The system shall allow users to add remarks or notes for each subject
- The system shall allow users to plan a semester course by selecting a degree program and semester
- The system shall allow users to input the courses, available professors and lecturers planned for the semester
- The system shall provide an option to indicate subjects that are offered more or less often due to student volume
- The system shall allow users to input the planned number of weekly hours according to the StuPo and the schedule at the start of the semester
- The system shall allow users to input the actual number of weekly hours at the end of the semester
- The system shall allow users to create services, including selecting the subject with the corresponding number, name and faculty
- The system shall allow users to specify which faculty benefits from the service
- The system shall enable the user to set the planned number of weekly hours for each subject based on the StuPo and schedule
- The system shall allow users to add remarks for each service

- The system shall allow the user to plan services, including viewing and selecting all subjects, their corresponding data and lecturers
 - The system shall allow users to set the actual number of weekly hours for each subject based on the StuPo and schedule
 - The system shall allow the user to input the actual hours accounted for each lecturer, ensuring accurate tracking of their workload
 - The system shall provide the functionality for the user to enter remarks related to the services provided, allowing for additional information
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- The system shall display the weekly hours assigned to professors, including an overview of their credit hour account
 - The system shall allow users to adjust a professor's workload based on tutorials, labs, or shared teaching responsibilities
 - The system shall allow the user to view all professors functions or administrative roles
 - The system shall enable the user to enter the functions or roles that each professor had or currently holds
 - The system shall provide the capability to record and manage the load reduction amount for each professor
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- The system shall show and generate reports listing professors' planned weekly hours and assigned functions
 - The system shall show and generate reports listing lecturers' planned weekly hours, including contact information
 - The system shall show and generate reports listing services provided by other departments
 - The system shall show and generate reports listing services used by other departments
 - The system shall show and generate reports listing subjects broken down by degree program and semester
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- The system shall store the data from prior semesters for a few decades
 - The system shall provide the data from the old semester as a template when planning a new semester

3.2 Nonfunctional Requirements

- **Performance:** Data uploads and data exports (reports) must be processed quickly within a few minutes
- **Scalability:** The system must be able to store up to 100,000 records efficiently
- **Availability:** The system should maintain 99% uptime with backups to ensure data recovery
- **Data Integrity:** The system must ensure that data is consistent throughout all processes. This should shall be done by the use of constraints
- **Normalization:** The system must implement normalization rules to eliminate redundancy
- **Documentation:** The system must be thoroughly documented